

Changyu (Thomason) Zhao

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EDUCATION

University of Wisconsin - Madison

Bachelor of Science in Computer Science (Honors) and Math

Cumulative GPA: **3.97/4.0**, Dean's List 2021 - 2023

Sep. 2021 – May 2025

Madison, Wisconsin

WORK EXPERIENCE

Tencent Xuanwu Lab

Security Researcher Intern

Aug. 2023

Beijing, China

- Developed a **hardware-assisted dynamic control flow analysis** tool for Linux & Windows platform based on **Last Branch Record (LBR)**, an Intel hardware feature that records the branch jump addresses.
- Implemented **kernel modules/drivers** to enable LBR and collect branch records from the kernel.
- Designed a user-mode analysis library to **reconstruct the control flow graph** from collected branch records.

UW - Madison Cybersecurity Operation Center

Cybersecurity Student Intern Analyst

Feb. 2023 – Present

Madison, Wisconsin

- Managed security incidents, including respond to alerts, investigate security breaches, and take corrective action.
- Use **Elastic** and **Palo Alto Firewall** to monitor and **analyze campus network traffic**, to identify potential threats and vulnerabilities and notify security alert to related users or departments.
- Conducted **Identity and Access Management (IAM)** for access restrictive student employee accounts.
- Processed **Daliy Threat Intel** with team members, fostering a **collaborative environment** to address security concerns and mitigate potential threats in campus.

RESEARCH EXPERIENCE

ASU SEFCOM/CDF Security Research Apprenticeship

Research Assistant

May. 2023 – Aug. 2023

Tempe, Arizona

- Developed and implemented a security research tool named "**LibHopper**," a **Python** based **automated exploitation primitives finding framework** for arbitrary user libraries.
- Leveraged automated processes to pipeline the evaluation of security implications related to **memory corruption vulnerabilities** in library internal structures, enhancing the capabilities of security researchers and developers.
- Employed a multifaceted methodology involving **customizing debugger scripts** and **symbolic execution** techniques to conduct comprehensive exploitability analysis.
- Identified and cataloged **assets of exploitable primitives** within library internal states, derived from meticulous analysis of memory corruption scenarios and symbolic execution findings on 6 widely-used C standard libraries.

PROJECTS

Router automated vulnerability discovery | *IDA, GDB, QEMU, Python*

May 2022 – Aug. 2022

- Exploited unfixed Common Vulnerabilities and Exposures (CVE) in **Tenda AC18 router firmware**.
- Employed **qemu** to rehost arm32 router firmware on an x86-64 machine, enabling systematic debugging.
- Developed **automated** stack overflow and command injection detection IDA plug-in at decompiled code level.
- Achieved Remote Code Execution (RCE) by **exploiting** vulnerabilities identified by the IDA plug-in.
- Discovered multiple unfixed CVEs: CVE-2018-16334, CVE-2018-5767, CVE-2018-5768, CVE-2018-5770, CVE-2020-15916, CVE-2022-28556, CVE-2022-28557.

EXTRACURRICULAR ACTIVITIES

r3kapiG CTF Team

Team member

Sep. 2021 – Present

- Achieved a notable accomplishment by securing a rank 12/1282 worldwide in the DEFCON 30 CTF Finals (2022), and 7/1828 worldwide in the DEFCON 31 CTF Finals (2023).
- Participated in over 40 48-hour Capture the Flag (CTF) competitions since 2021, with a focus on Binary Exploitation (PWN) and Reverse Engineering challenges.

- Attained the Yellow Belt and Blue Belt levels by completing all 472 challenges in the Dojo instance, currently ranked 43 out of 2500+ participants.
- Acquired comprehensive skills in various areas of binary exploitation, including program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race conditions, and kernel security.

TECHNICAL SKILLS

Proficient in reverse engineering, have reverse experience in large-scale projects. **Familiar** with control flow tracking, data flow tracking, symbolic execution and the use of binary analysis tools.

Programming Languages: C/C++, Python, Java, x86-Assembly, JavaScript, HTML/CSS, $\text{\LaTeX}$

Binary Security Tools: IDA/Ghidra, GDB/Windbg, Angr, Frida, Joern

Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux

Soft Skills: Time/Project Management, Collaboration, Conflict Resolution, Active Listening